

Problem Set 5

Due on Thursday August 19 at 11:59 PM (if you want feedback)

1. Find $F'(x)$ for each of the following. (You do not need to evaluate any integrals.)

(a) $F(x) = \int_0^1 \sqrt{x^2 + t^4} dt$

(b) $F(x) = \int_{x^{\frac{1}{3}}}^1 e^{-xt^3} dt$

2. Solve each of the following initial-value problems.

(a) $y'(t) = (y(t))^2; \quad y(0) = \frac{1}{2}$

(b) $y'(t) = 2y(t) + t; \quad y(0) = 3$

(c) $y'(t) + 2ty(t) = e^{-t^2}; \quad y(0) = -1$

(d) $y'(t) = \frac{5}{t}y(t) + t^{10}; \quad y(1) = 7$

(e) $y'(t) = t^2(1 + y(t)); \quad y(0) = 1$

3. Solve the integral equation

$$y(t) + 4 \int_0^t y(s) ds = t + 3.$$

4. Solve the initial-value problem

$$y'(t) + y(t) = y(t)^3; \quad y(0) = 2.$$

(Suggestion: Make the substitution $u(t) = y(t)^{-2}$.)

5. Find the general solution of each of the following:

(a) $y''(t) - y'(t) + \frac{1}{4}y(t) = 0$

(b) $y''(t) + 2y'(t) - 15y(t) = 0$

(c) $t^2y''(t) + 3ty'(t) + y(t) = 0, \quad t > 0$

6. In the framework of the Black-Scholes model, consider a stock with volatility $\sigma = .20$ and assume that the interest rate is $r = .02$. The current price of the stock is x (with $2,800 < x < 3,500$). Let τ be the first future time at which the stock price hits either 3,500 or 2,800. We have an option that pays 100 if the stock price is 3,500 at τ and pays nothing if the stock price is 2,800 at τ . (The option also pays nothing if the stock price stays strictly between 2,800 and 3,500 forever.) The price of this option satisfies the boundary value problem

$$\frac{1}{2}\sigma^2x^2v''(x) + rxv'(x) - rv(x) = 0, \quad v(2,800) = 0, \quad v(3,500) = 100.$$

Find a formula for $v(x)$ and compute $v(3,100)$.